

Week 3 - part 3

CS 4200-6200

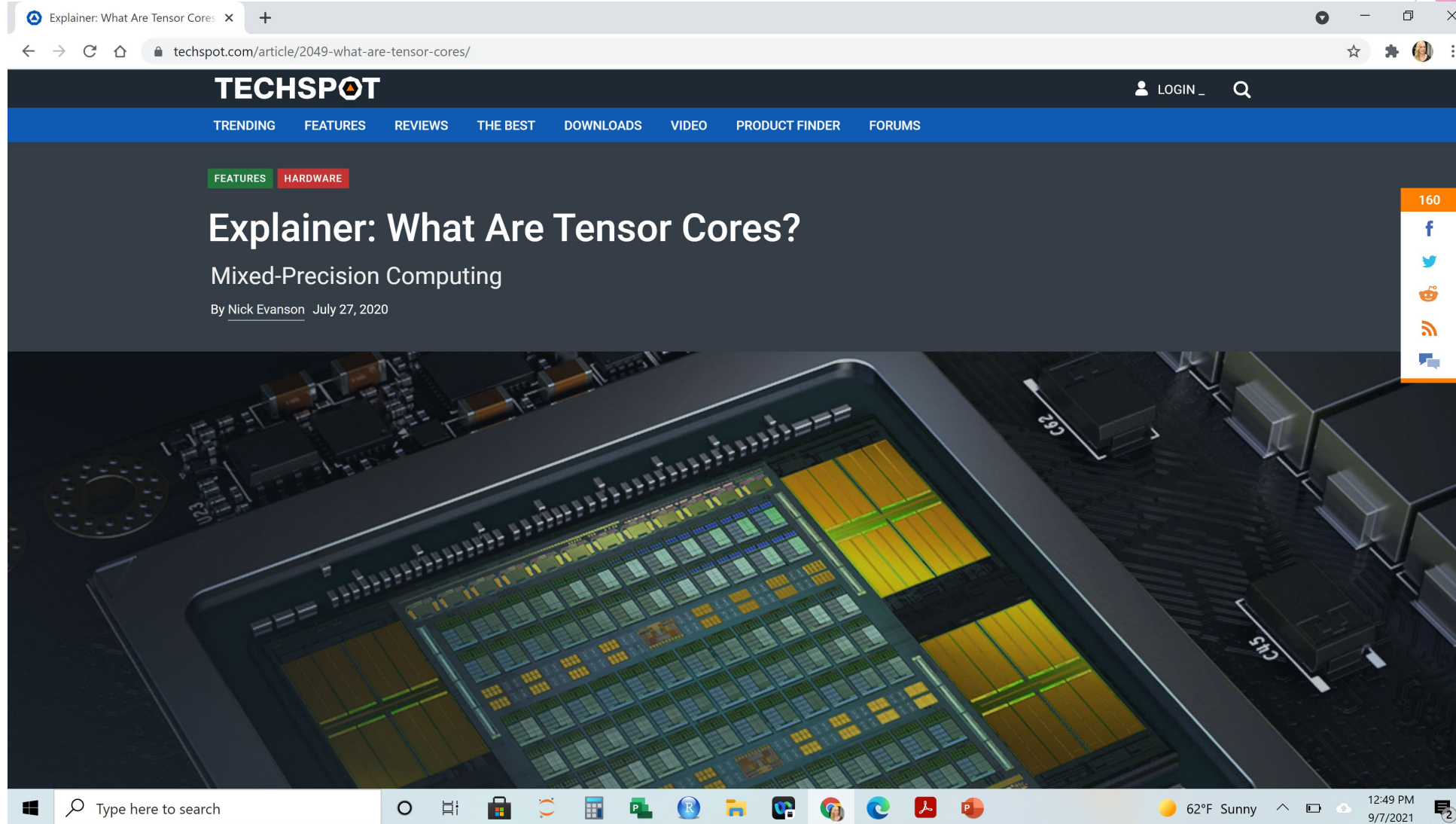
Linear Algebra for QC

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Week 3

Reminder of Agenda / where we are

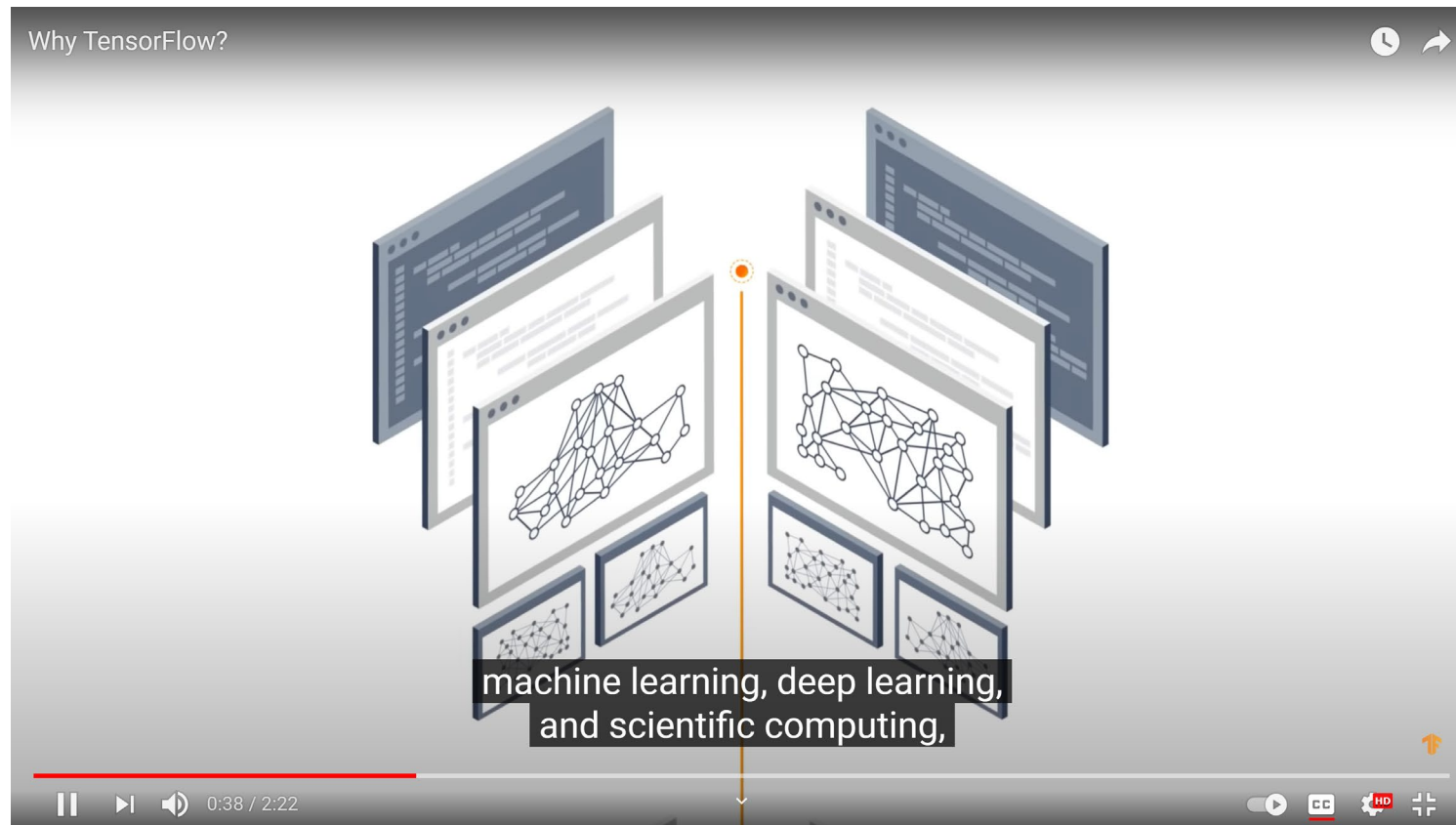
- ▶ We covered BIG PICTURE of Quantum Weeks 1 & 2
- ▶ We also covered very top level of Physics of Quantum Mechanics
- ▶ In this week, we will cover Linear Algebra (may be review for some)
 - ▶ This is so we can better understand our Quantum Logic
- ▶ First Video Lecture (W3-part 1) will be Why use Matrices, Matrix Addition, and Scalar Multiplication
- ▶ Previous Video Lecture (W3-part 2) and Multiplying Two Matrices, Naïve method, Strassen Method, and other more efficient method for large matrices
- ▶ This, the final Video Lecture (W3-part 3) will be more about Dirac Notation, How Linear Algebra Describes Quantum Gates, and Tensors

Tensor Cores - GPU - Graphics



TensorFlow

<https://www.youtube.com/watch?v=yjprp0oH5c8>



What is a Tensors

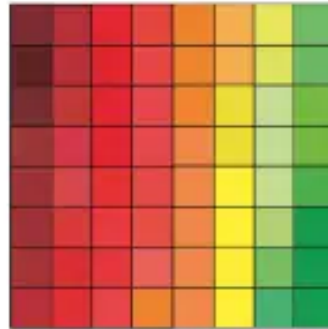
tensor = multidimensional array

vector



$$\mathbf{v} \in \mathbb{R}^{64}$$

matrix



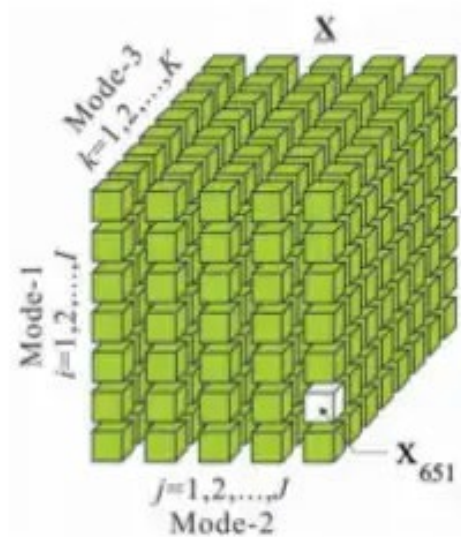
$$\mathbf{X} \in \mathbb{R}^{8 \times 8}$$

tensor



$$\mathbf{X} \in \mathbb{R}^{4 \times 4 \times 4}$$

third-order tensors

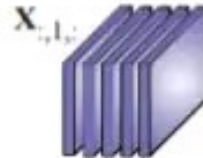


$$\mathcal{X} \in \mathbb{R}^{7 \times 5 \times 8}$$

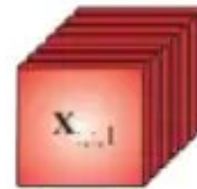
Horizontal Slices



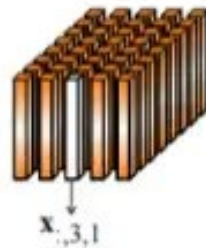
Lateral Slices



Frontal Slices



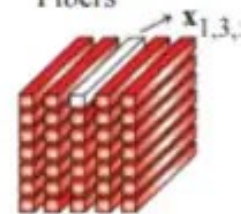
Column (Mode-1)
Fibers



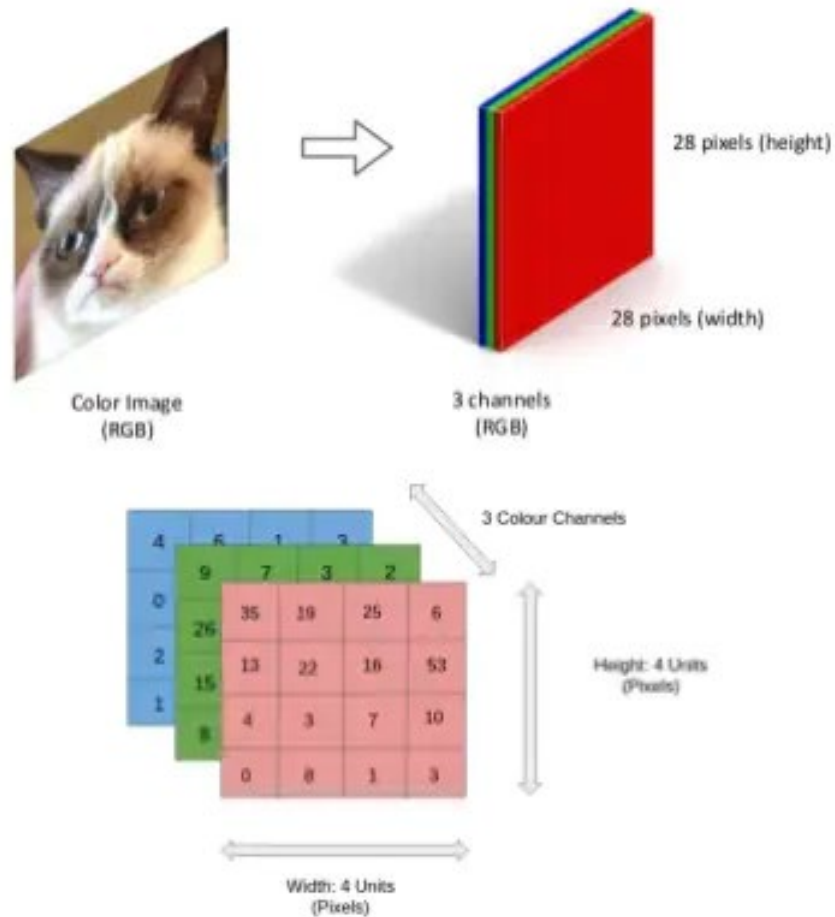
Row (Mode-2)
Fibers



Tube (Mode-3)
Fibers



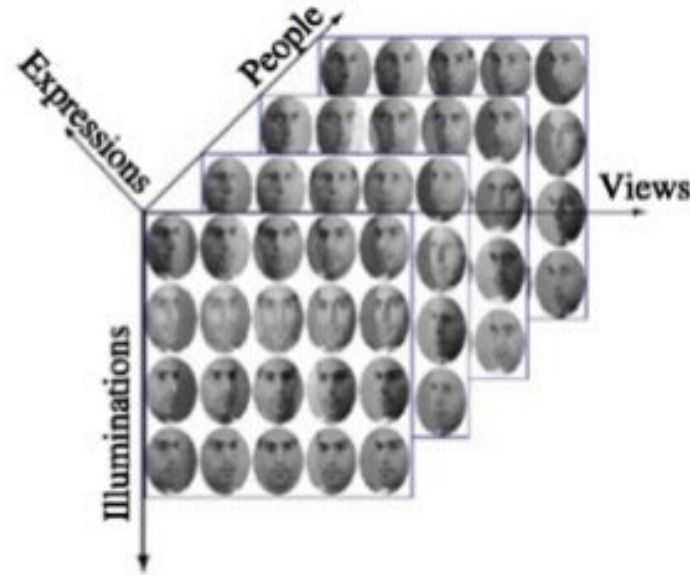
color image is 3rd-order tensor



color video is 4th-order tensor



facial images database is 6th-order tensor



Two Handle Affects between two Qubits we will use Kronecker Product

$$\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \otimes \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix} = \begin{bmatrix} a_{1,1} \begin{bmatrix} b_{1,1} & b_{1,2} \end{bmatrix} & a_{1,2} \begin{bmatrix} b_{1,1} & b_{1,2} \end{bmatrix} \\ a_{2,1} \begin{bmatrix} b_{1,1} & b_{1,2} \end{bmatrix} & a_{2,2} \begin{bmatrix} b_{1,1} & b_{1,2} \end{bmatrix} \end{bmatrix} = \begin{bmatrix} a_{1,1}b_{1,1} & a_{1,1}b_{1,2} & a_{1,2}b_{1,1} & a_{1,2}b_{1,2} \\ a_{1,1}b_{2,1} & a_{1,1}b_{2,2} & a_{1,2}b_{2,1} & a_{1,2}b_{2,2} \\ a_{2,1}b_{1,1} & a_{2,1}b_{1,2} & a_{2,2}b_{1,1} & a_{2,2}b_{1,2} \\ a_{2,1}b_{2,1} & a_{2,1}b_{2,2} & a_{2,2}b_{2,1} & a_{2,2}b_{2,2} \end{bmatrix}.$$

Table B.2. 3-3: R_{gate} correlation matrices

1x2 case	2x2 case
$R_{\text{gate}} = R_{\text{UG}} = \begin{bmatrix} 1 & \beta \\ \beta^* & 1 \end{bmatrix}$	$R_{\text{gate}} = R_{\text{dUG}} \otimes R_{\text{UG}} = \begin{bmatrix} 1 & \alpha \\ \alpha^* & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & \beta \\ \beta^* & 1 \end{bmatrix} = \begin{bmatrix} 1 & \beta & \alpha & \alpha\beta \\ \beta^* & 1 & \alpha\beta^* & \alpha \\ \alpha^* & \alpha^*\beta & 1 & \beta \\ \alpha^*\beta^* & \alpha^* & \beta^* & 1 \end{bmatrix}$

Next Week (Week 4) we will look at Quantum Logic

- ▶ Several Rules apply to Quantum
- ▶ Such as No Cloning Theorem
- ▶ If anyone or anything can deduce a state, the state collapses
- ▶ All operations have to be Unitary - they can be done but they also can be undone
- ▶ We have real-life problems of Decoherence and Interference
- ▶ We have real-life problems of dealing with Error and Error Correction
- ▶ Next week we will look at DWQ - Chapter 6 - “What do you mean Probably”